Log Analytics Workspace in Azure Data Factory

A practical learning guide covering purpose, architecture, monitoring, KQL queries, setup, and the Edukron project use case

1. What is a Log Analytics Workspace?

Azure Log Analytics Workspace (LAW) is a centralized service in Azure Monitor where diagnostic logs and telemetry from Azure resources can be collected, stored, and queried. In an Azure Data Factory (ADF) solution, it can store information about pipeline runs, activity runs, and trigger runs.

Think of it as a searchable logbook for your data factory. ADF executes pipelines; diagnostic settings send selected execution logs to the workspace; you then use Kusto Query Language (KQL) to investigate what happened and identify patterns.

Important: Creating a Log Analytics Workspace does not automatically send ADF logs to it. You must configure Diagnostic settings on the ADF resource and select the workspace as a destination.

2. Why is it used with Azure Data Factory?

Use case	How it helps
Failure investigation	Find failed pipeline or activity runs and inspect the reported error details.
Performance monitoring	Track pipeline duration and identify activities that take unusually long.
Historical analysis	Query stored run information over a selected time range, rather than relying only on the curre
Operational visibility	Review pipeline, activity, and trigger execution patterns in one queryable location.
Alerts and reporting	Build Azure Monitor alerts or workbooks based on log data and operational conditions.

3. How does ADF send logs to Log Analytics?

Flow: Azure Data Factory → Diagnostic Settings → Log Analytics Workspace → KQL queries / Azure Monitor alerts

Azure Data Factory	**Diagnostic Settings**	**Log Analytics Workspace**	**Query & alert**
Runs pipelines, activities and triggers	Selects log categories and destination	Stores the selected diagnostic data	KQL, workbooks and Azure Monitor

You choose which diagnostic categories to export. Common ADF categories include **PipelineRuns**, **ActivityRuns**, and **TriggerRuns**. Once configured, ADF sends the selected diagnostic records to the chosen workspace.

4. ADF Monitor vs. Log Analytics Workspace

ADF Monitor	Log Analytics Workspace
Best for checking recent pipeline and activity execution status.	Best for querying and analyzing diagnostic logs over time.
Provides a convenient visual view of run status, duration, inputs, outputs, and errors.	Provides KQL-based filtering, aggregation, and analysis, and inte
Useful for day-to-day operational troubleshooting.	Useful for deeper investigation, historical analysis, and custom mo

Remember: ADF Monitor is the built-in execution-monitoring interface. Log Analytics is the centralized log store and query environment. They complement each other; one does not replace the other.

5. Querying ADF logs using Kusto Query Language (KQL)

KQL is the query language used by Azure Monitor Logs. After ADF diagnostic logging is configured, you can query the relevant tables in the Log Analytics workspace.

Example A — View pipeline runs from the last 24 hours

```
ADFPipelineRun
| where TimeGenerated > ago(24h)
| project PipelineName, Status, RunId, Start, End
| order by TimeGenerated desc
```

Example B — Find failed pipeline runs

```
ADFPipelineRun
| where Status == "Failed"
| project PipelineName, RunId, Status, TimeGenerated
| order by TimeGenerated desc
```

Example C — Find failed activities and their errors

```
ADFActivityRun
| where Status == "Failed"
| project PipelineName, ActivityName, Error, TimeGenerated
| order by TimeGenerated desc
```

Table-name note: The examples use resource-specific table names such as ADFPipelineRun and ADFActivityRun. These are available when the diagnostic setting uses the resource-specific destination mode. If you use the legacy Azure Diagnostics mode, records may instead appear in the **AzureDiagnostics** table, with category information in columns. The exact available columns depend on the diagnostic destination mode and configuration.

6. How to configure it and how it fits the Edukron project

1. **Open the ADF resource:** In the Azure portal, open your Data Factory resource.

2. **Open Diagnostic settings:** Under the Monitoring section, select *Diagnostic settings*.

3. **Add a diagnostic setting:** Select *Add diagnostic setting* and provide a meaningful name.

4. **Select log categories:** Enable the categories you need, such as PipelineRuns, ActivityRuns, and TriggerRuns.

5. **Choose the destination:** Select *Send to Log Analytics workspace*, then choose an existing workspace or create one.

6. **Save:** Save the diagnostic setting. New matching diagnostic events will be sent to the workspace.

7. **Validate:** Run an ADF pipeline, wait for diagnostic data to arrive, then open the workspace and test a KQL query.

Edukron project recommendation
Your simplified Edukron retail pipeline uses ADF to trigger the existing Databricks Workflow Job. After the ADF pipeline is successfully triggering that job, configure a Log Analytics Workspace for ADF diagnostics. Enable PipelineRuns, ActivityRuns, and TriggerRuns so you can monitor whether the ADF trigger pipeline succeeded, how long its activities took, and whether an ADF activity failed.

Where should you investigate a failure?

• **ADF pipeline or activity failure:** Start in ADF Monitor to identify the failed activity and inspect its error output. Use Log Analytics for KQL-based history and trends.

• **Databricks notebook/job failure:** Open the Databricks Workflow Job run and inspect the failed task, notebook output, driver logs, and Spark execution details. ADF logs may show that the Databricks call failed, but the detailed notebook exception is usually in the Databricks run itself.

Quick revision: ADF executes the pipeline → Diagnostic Settings export selected logs → Log Analytics stores them → KQL helps investigate → Azure Monitor can alert on important conditions.

Learning note: Log Analytics Workspace is part of Azure Monitor. It is not a data lake and it is not where your retail Bronze/Silver/Gold datasets should be stored. Keep business data in ADLS Gen2 and use Log Analytics for operational monitoring and diagnostics.